

WASP-10b

R-0806

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_181102 · created 2026-07-28T11:13:01+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2458424.7916 +/- 0.0015 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.173 +/- 0.0089
Transit depth (Rp/Rs)^2	2.99 +/- 0.31 [%]
Orbital Inclination (inc)	87.83 +/- 0.64
Airmass coefficient 1 (a1)	1.17 +/- 0.015
Airmass coefficient 2 (a2)	-0.0976 +/- 0.0035
Scatter in the residuals of the lightcurve fit is	1.31 %
Best Comparison Star	None
Optimal Aperture	4.39
Optimal Annulus	13.03
Transit Duration (day)	0.0891 +/- 0.0041

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R $\blacksquare$ fit vs published	0.173 $\pm$ 0.0089 vs 0.1592 (deviation 1.29 σ)
Duration fit vs published	0.0891 d vs 0.0946 d (deviation 1.11 σ)
Mid-time O-C	-5.4 min
Depth SNR	16.2
Residual scatter	1.31 %

Light curve

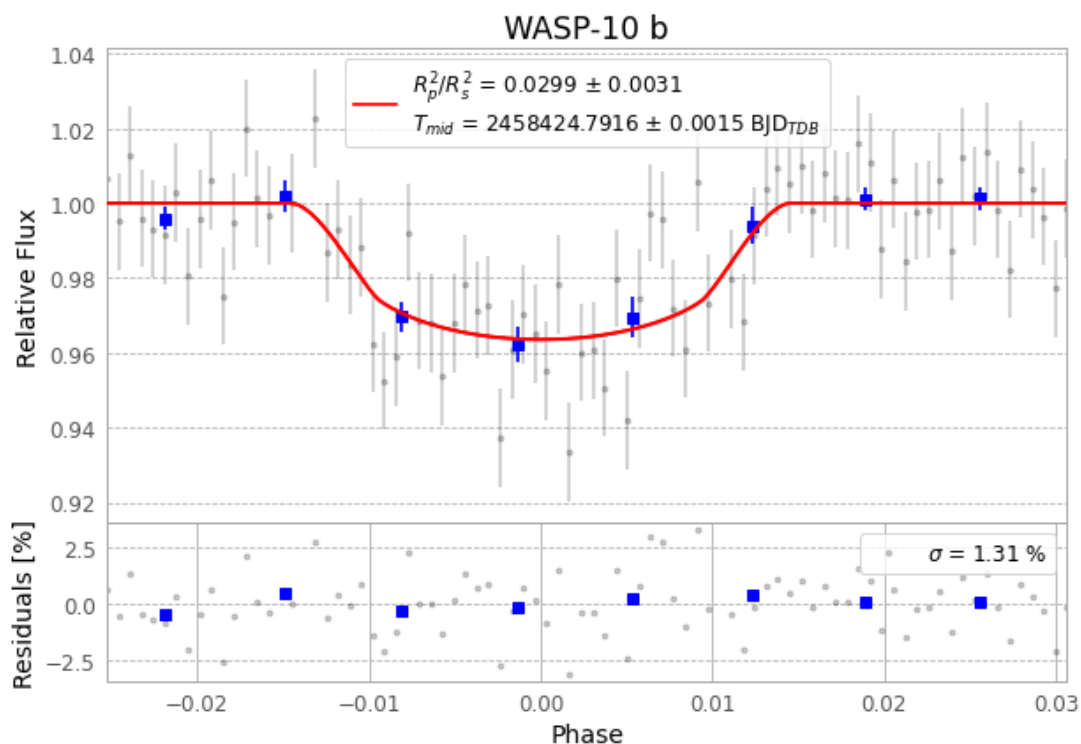


Figure 1 — FinalLightCurve_WASP-10 b_2018-11-01.png (SHA-256 73f5f18b7af2e7e1...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [297, 325], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 bf50c32c0362e4bd...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

84 FITS frames (55 MB), WASP-10181102050012.FITS → WASP-10181102090912.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-181102.log (6683c1a163be...), FinalLightCurve_WASP-10 b_2018-11-01.png (73f5f18b7af2...), FinalLightCurve_WASP-10 b_2018-11-01.csv (9cc9e6bbe220...)

Compute resources

Wall time 130.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 11:13Z.